



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Supplementary Examinations, July-2025

DISCRETE MATHEMATICS

(COMMON TO CSE, IT, CSE (AI&ML), CSE (DS))

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

- Define functionally complete set of connectives with examples
- Construct truth table for $(P \rightarrow (Q \vee R)) \vee S$
- Let $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) | x > y\}$. Draw the graph of R.
- Let $f: R \rightarrow R$ and $g: R \rightarrow R$, where R is the set of real numbers. Find a) $f \circ g$

(10 Marks)

[1M] L1

[1M] L1

[1M] L2

[1M] L1

[1M] L2

[1M] L2

- Where $f(x) = x^2 - 2$, $g(x) = x + 4$.
- Let Z^+ is the set of positive integers. Then prove that $(Z^+, +)$ is semi group.

f) Represent the Hasse diagram representing the positive divisors of 36.

g) How many 5 card hands have 2 cards of one suit and 3 cards of different suit from a deck of 52 cards?

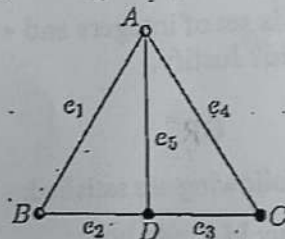
[1M] L1

[1M] L3

[1M] L1

h) Determine the coefficient of $x^4 y^7$ in the expansion of $(y - 3x)^{11}$.

i) Write the Euler path for the following graph.



- Explain the Hamilton graph with example.

[1M] L2

PART- B

(5 X 10 = 50 Marks)

- Verify which of the following are tautologies and contradictions.

[10M] L3

a) $(P \rightarrow (Q \vee R)) \wedge (\sim Q) \wedge \sim (P \rightarrow R)$.

b) $[(P \rightarrow \sim Q) \wedge (\sim R \rightarrow S) \wedge (P \vee \sim R)] \rightarrow (\sim Q \vee S)$

With or without using truth table.

OR

3. Verify that the following argument is valid.

[10M] L3

If Joe is a mathematician, then he is ambitious.

If Joe is an early riser, then he does not like oatmeal. If Joe is ambitious, then he is an early riser.

Hence, if Joe is a mathematician, then he does not like oatmeal

4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$, where $\mathbb{R}$ is the set of real numbers. Find $f \circ g$, $g \circ f$, where $f(x) = x^2 - 2$, $g(x) = x + 4$. State whether these functions are injective, surjective and bijective

[10M] L4

OR

5. Give an example of a relation which is

[10M] L3

- i) Neither reflexive nor irreflexive,
- ii) Both symmetric and anti-symmetric,
- iii) Each reflexive, symmetric and transitive,
- iv) Both symmetric and transitive but not reflexive

6. a) Prove that $(\mathbb{Z}, *)$, where $\mathbb{Z}$ is set of integers and $*$ is defined as $a * b = a + b - 1$ is an Abelian group.

[5M] L6

b) Prove that $(\mathbb{Z} - \{-1\}, *)$, where $\mathbb{Z}$ is set of integers and $*$ is defined as $a * b = a + b + 1$ is a commutative semigroup. Is it monoid? Justify.

[5M] L6

OR

7. Construct the examples of which the following are satisfied:

[10M] L5

- a) Semigroup but not monoid.
- b) Monoid but not group.
- c) Group but not Abelian group.
- d)

8. a) Determine the term containing $x^{10}y^5w^{10}z^5$ in the expansion of $(2x - 5y - 6z + 3w)^{30}$.

[5M] L6

b) Find the number of non-negative integral solutions satisfying $x_1 + x_2 + x_3 < 15$.

[5M] L6

OR

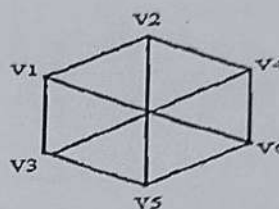
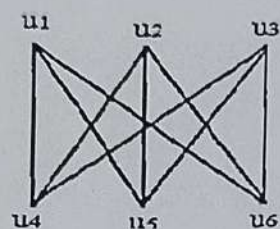
9. Determine the number of arrangements of the letters in TALLAHASSEE

[10M] L6

- a) which have no adjacent A's.
- b) where the letters S are together and the letters E are together.
- c) begin with E and end with S.

10. Prove that the following two graphs are isomorphic.

[10M] L6

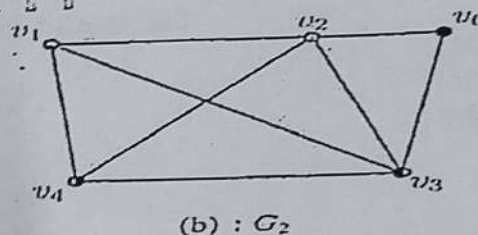
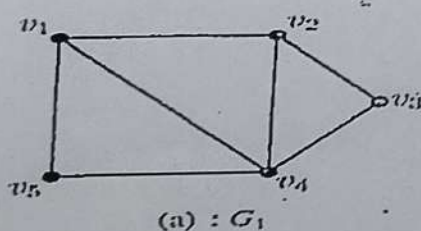


OR

11. Consider the following two graphs G_1, G_2 . Then obtain the following.

[10M] L6

- $G_1 \cup G_2$
- $G_1 \cap G_2$
- $G_1 \Delta G_2$
- $\bar{G}_1 \cup \bar{G}_2$
- $\bar{G}_1 \cap \bar{G}_2$



---ooOoo---



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, January-2025

DISCRETE MATHEMATICS

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) Find CNF of $P \vee (P \rightarrow Q)$ [1M] L2
- b) Define Tautology. [1M] L1
- c) If $X = \{1, 2, 3, 4\}$ and the relation $R = \{(x, y) / x < y\}$. Draw the graph of 'R'. [1M] L2
- d) Define Partial ordering. [1M] L1
- e) Find the order of an elements i and $-i$ of multiplicative group $G = \{1, -1, i, -i\}$ [1M] L1
- f) Define index of a subgroup of a finite group. [1M] L1
- g) How many words of three distinct letters can be formed from CAKE? [1M] L1
- h) Define permutation and give an example. [1M] L1
- i) Draw the graph $K_{3,3}$ [1M] L1
- j) Define Spanning tree. [1M] L1

PART- B

(5 X 10 = 50 Marks)

2. a) Find DNF of $(\sim P \rightarrow R) \wedge (P \leftrightarrow Q)$ [5M] L3
- b) Verify the validity of the following argument

Every living thing is a plant or animal

John's gold fish is alive and it is not a plant

All animals have hearts

Therefore john's gold fish has a heart

[5M] L3

OR

3. Show that $P \rightarrow (Q \wedge R)$ and $\sim P \vee (Q \wedge R)$ are logically equivalent by using truth tables. [10M] L4

4. a) Let $P(A)$ be the power set of any non empty set A , then prove that the

relation $\subseteq$ of inclusion is not an equivalence relation

[5M] L5

- b) Let R be a relation on set $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$.

Find the reflexive, symmetric and transitive closure of R .

[5M] L2

OR

5. a) Prove that a function $f: R \rightarrow R$ defined by $f(x) = 3x - 7$ is a bijective. [5M] L4

- b) Draw the Hasse diagram for the positive divisors of 72. [5M] L2

6. Show that fourth root of unity forms a group under multiplication

[10M] L3

OR

7. a) If a, b are any two elements of a group $(G, \cdot)$ which commute. Show that a^{-1} and b commute.

[5M] L4

b) Prove that identity element of a group G is unique.

[5M] L5

8. a) Find the number of integers between 1 to 300 that are divisible by at least one of 3, 5, 7

[5M] L3

b) Find the coefficient of x^7y^5 in the expansion of $(2x - 3y)^{12}$

[5M] L5

OR

9. a) Prove that $C(n, r) C(r, k) = C(n, k) C(n - k, r - k)$ for integers $n \geq r \geq k \geq 0$

[5M] L5

b) Using multinomial theorem, expand $(2x - 3y + 4z)^3$

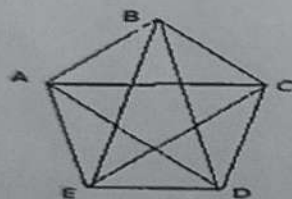
[5M] L3

10. a) Write the adjacency matrix and incidence matrix of the graph whose adjacency list is given below.

Vertices	Adjacent vertices
a	b, d
b	a, c, c
c	b, e, d
d	a, c
e	b, c

[5M] L6

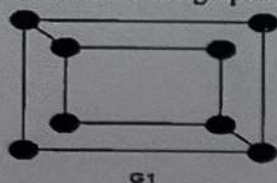
b) Find the Euler path or Euler circuit if it exist, if not explain why?



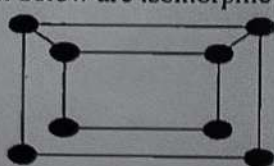
[5M] L2

OR

11. Determine whether the graphs given below are isomorphic?



G1



G2

[10M] L3

---ooOoo---



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, January-2025

DISCRETE MATHEMATICS

(COMMON TO IT, CSE (AI&ML), CSE (DS))

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) Translate the sentence into symbolic form of "Some men are gaint" [1M] L2
- b) Define Truth table. [1M] L1
- c) If $f(x) = x + a$ and $g(x) = x - a$ then find $gof(x)$ [1M] L1
- d) Give an example of a relation which is transitive and reflexive but not symmetric [1M] L1
- e) Define Monoid [1M] L1
- f) Show that binary operation $*$ defined on $(R, *)$ by $x * y = x^y$ is not an associative. [1M] L1
- g) In how many ways can all the letters in "MATHEMATICAL" is arranged. [1M] L1
- h) Find the number of positive integer solutions of the equation $x_1 + x_2 + x_3 = 14$ [1M] L1
- i) Define Planar Graph [1M] L1
- j) Draw the complete graph K_5 [1M] L1

PART- B

(5 X 10 =50 Marks)

2. Obtain PCNF of $(\sim P \rightarrow R) \wedge (P \rightarrow Q) \wedge (Q \rightarrow P)$

[10M] L3

OR

3. a) Verify the validity of the following arguments

Tigers are dangerous animals

There are tigers

There are dangerous animals

[5M] L3

- b) Derive $\sim P$ by using indirect method from the following premises

$(R \rightarrow \sim Q), R \vee S, S \rightarrow \sim Q, P \rightarrow Q$

[5M] L4

4. a) Show that the relation "greater than or equal to" is a partial ordering on the set of integers. [5M] L4

- b) If $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) / x < y\}$. Draw the graph of 'R' and also give its matrix. [5M] L4

OR

- P
5. a) Draw the Hasse diagram $(P(A), \subseteq)$ where $A = \{a, b, c\}$ [5M] L2
 b) Prove that the relation R defined by "a is congruent to b modulo m" on the set of integers is an equivalence relation [5M] L2
6. If $(G, *)$ is an abelian group iff $(a * b)^2 = a^2 * b^2$ for all $a, b \in G$ [10M] L2
 OR
7. Show that the set Q^+ , of all +ve rational numbers forms an abelian group under the composition defined by by "O" such that $aOb = \frac{ab}{3}$ for $a, b \in Q^+$. [10M] L4
8. a) Find the number of integers between 1 and 250 both inclusive that are divisible by any of the integers 2, 3, 5, 7. [5M] L5
 b) Expand the multinomial $(x_1 + x_2 + x_3 + x_4)^4$ [5M] L3
 OR
9. a) A certain computer centre employs 80 computer programmers of these 37 can program PYTHON and 26 in JAVA and 18 can program in both languages. How many can program in neither of these 2 languages? [5M] L6
 b) Find the number of ways so that the 8 boys and 8 girls to sit around a circular table, if boys and girls sit alternatively. [5M] L5
10. a) State and prove hand shaking property. [5M] L3
 b) Define Chromatic number of a graph. Explain it through an example. [5M] L4
 OR
11. a) Give an example graph which is Hamiltonian but not Eulerian with explanation. [5M] L3
 b) Give an example graph which is Eulerian but not Hamiltonian with explanation. [5M] L3

---ooOoo---